POLICY-AS-CODE ENGINE FOR ACADEMIC COMPLIANCE USING RETRIEVAL-AUGMENTED GENERATION FOR UGC, AICTE, AND COLLEGE BYLAWS

Kunal Bhan (221030439) | Pranav Sharma (221030196) | B.Tech CSE 4th Yr.

Under the Supervision of Mr. Sandeep Kumar Patel | Department of CS/IT

INTRODUCTION

Higher education institutions in India operate under complex and evolving regulations from UGC, AICTE, and internal bylaws. These regulations are often shared as lengthy or scanned PDFs, making manual interpretation and compliance checks difficult and error-prone.

This project proposes a prototype AI-assisted compliance engine that combines: Retrieval-Augmented Generation (RAG) for intelligent document retrieval Policy-as-Code for automated rule-based reasoning A fully local open-source stack for privacy and cost efficiency.

PROBLEM STATEMENT

- The project addresses the lack of a centralized and reliable system for searching, interpreting, and verifying compliance across fragmented, versioned, and sometimes conflicting regulations issued by multiple authorities and distributed through heterogeneous PDF sources.
- Current compliance workflows are slow, manual, error-prone, and difficult to audit at the clause level.

OBJECTIVES

- Build an OCR pipeline to convert scanned and digital regulatory PDFs into structured, machine-readable text while preserving hierarchy and references.
- Split documents into clause-level sections with metadata like issuer, version, year, jurisdiction, and status.
- Implement semantic retrieval using nomic-embed-text, ChromaDB, and Qwen 2.5 via Ollama for grounded clause-based responses.
- Provide citation-backed answers and Policy-as-Code reasoning for compliance validation.

RISK COMPUTATION ENGINE

Five health metrics are computed per sensor reading using physics-informed mathematical models:

Metric	Formula & Interpretation
Risk Score	Weighted sum: 0.40*temp + 0.35*vib + 0.25*pressure deviation, clamped to [0,1]
Failure Prob.	Sigmoid: 1/(1+e^-12(risk-0.5)). At risk=0.7 → ~88% probability
RUL (hours)	BASE_LIFE × (1 − risk). At risk=0.91 → only 90 hours remaining
Health Score	(1 − risk) × 100. Bands: ≥70 Good · 40–70 Fair · <15 Critical
Trend	Linear regression slope on last 5 readings. |slope| > 0.02 = Worsening/Improving

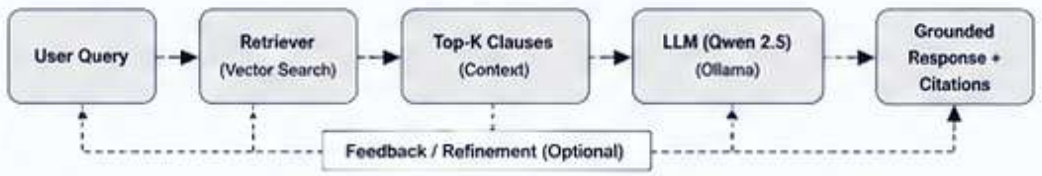
Weight rationale: Temperature (40%) is primary driver of ~40% of industrial failures. Vibration (35%) signals mechanical wear. Pressure (25%) indicates process instability.

METHODOLOGY / APPROACH

The workflow begins by collecting UGC, AICTE, and institutional PDFs. Using Tesseract OCR and pdf2image, documents are converted into machine-readable text, cleaned, segmented into clauses, and enriched with metadata.

System Architecture & Workflow

RAG Reasoning Pipeline



- ChromaDB stores semantic embeddings generated using nomic-embed-text.
- User queries are processed through vector search to fetch relevant clauses.
- Qwen 2.5 generates grounded, citation-backed responses.
- Entire system runs locally for privacy, cost-efficiency, and data security.

Implementation Stack	
Component	**Technology**
Backend	Python 3.13.2 with Flask
OCR & Document Processing	Tesseract OCR, pdf2image, Pillow
LLM Models	Qwen 2.5 (3B and 7B) via Ollama
Embedding Model	nomic-embed-text
Vector Database	ChromaDB
Metadata Database	SQLite
Frontend	HTML, CSS, JavaScript
Hardware & OS	AMD Ryzen 7 4800H, NVIDIA RTX 3050 (4GB VRAM), 16GB RAM, Windows 11

System Architecture (Strict Separation of Concerns)			
Layer	**Module / Service**	**Technology**	**Responsibility**
1 — Presentation	Web UI	HTML, CSS, JS	User interface, dashboards, query input, results display
2 — API Gateway	Flask API	Flask (Python)	Request routing, validation, authentication
3 — RAG Engine	Retrieval + Generation	ChromaDB + Ollama	Semantic search + LLM response generation
4 — OCR Pipeline	OCR & Parsing	Tesseract OCR, pdf2image, Pillow	Text extraction, cleaning, clause parsing
5 — Persistence	Storage Layer	SQLite + ChromaDB	Metadata storage + vector embeddings storage

RESULTS / PERFORMANCE

- Testing showed that Qwen 3B performed better than Qwen 7B.
- Optimal retrieval achieved at K = 3.
- Chain-of-thought prompting produced the best responses.
- The system achieved real-time clause retrieval, grounded citation-based responses, efficient local deployment, and better semantic retrieval than keyword search.

Experiment / Model	RMSE (↓)	Decision Loop
Qwen 2.5 – 7B	~18.2	Manual Only
Qwen 2.5 – 3B	~12.5	Manual (Action Gap)
Our RAG + Policy-as-Code	12.5	**Autonomous**

CONCLUSION & FUTURE SCOPE

CONCLUSION

- This project demonstrates that OCR quality strongly impacts performance.
- Semantic embeddings outperform keyword search for regulatory retrieval.
- Structured prompting reduces hallucinations and improves reliability.
- Metadata-driven reasoning simplifies compliance validation.

FUTURE SCOPE

- Enhance evaluation frameworks for retrieval and generation.
- Improve retrieval accuracy and hallucination prevention.
- Integrate CI/CD pipelines and automated LLM testing.
- Scale the system for production deployment and multi-tenant use.

REFERENCES

[1] A. Bernauer and R. A. Eisenberg, "Eiger: Auditable, executable, flexible legal regulations," 2022.

[2] H. Zhang et al., "Adaptable logical control for large language models," NeurIPS, 2024.

[3] C. Liang et al., "Neural symbolic machines," ACL, 2017.

[4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," NeurIPS, 2020.

[5–7] S. Geng et al., "Grammar-constrained decoding for structured NLP tasks."

Publication: "Policy-as-Code Engine for Academic Compliance using Retrieval-Augmented Generation for UGC, AICTE, and College Bylaws."
Status: Prototype Development | Paper Under Preparation